

AI Voice Calling System Audit Report

Prophunt Real Estate — Internal Engineering Review

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Classification

Confidential — Internal Use Only

This report covers the full engineering audit of Prophunt's AI voice calling stack performed June 4–5, 2026: cost benchmarking against Ravan.ai Agni, latency analysis, live bug fixes applied to the production codebase, and a phased optimisation roadmap.

1. Executive Summary

Prophunt launched its AI outbound calling system on **May 14, 2026**. As of June 5, 2026, the system has processed approximately **1,170 TTS calls** and **667 STT requests** — pilot scale only. Total spend to date is negligible (~\$15–35 on ElevenLabs). The system is cost-appropriate for current volume.

Competitor Ravan.ai (Agni product) charges **■4/min** API or **■8/min** white-label. Prophunt's current internal cost at production scale (Sarvam STT + ElevenLabs TTS + GPT-4o-mini) is approximately **■3.5–4.5/min**, on par with Ravan.ai API pricing. There is clear headroom to reach a **■3/min internal cost** and a **■5–6/min sell price** through the phased roadmap in Section 6.

Metric	Current State	Target	Status
Internal cost / min	■3.5–4.5	■3.00	LabelYellow
End-user sell price	Not yet set	■5–6	LabelYellow
Ravan.ai API price	■4/min (ref)	—	—
Call latency P50	~1.8s	<1.5s	LabelYellow
Call latency P90	~2.5s	<2.0s	LabelRed
ElevenLabs TTS TTFB P50	110–155 ms	<200 ms	LabelGreen
Audio quality (crackling)	FIXED June 5	Clean	LabelGreen
Orchestrator stability	FIXED June 5	No crashes	LabelGreen

Table 1 — Executive summary scorecard

2. System Architecture

The system is a Node.js orchestrator deployed on **Railway (Southeast Asia)**, fronted by a Vercel dashboard. Audio flows through EnableX telephony via WebSocket.

2.1 Call Flow

Step	Component	Technology	Latency (P50)
1	Outbound dial	EnableX Voice API	~400 ms
2	Caller speech	EnableX WebSocket → μ-law 8 kHz	real-time
3	Speech-to-Text	Sarvam Saarika v2.5 (batch REST)	900–2 025 ms
4	Language detection	Orchestrator heuristics	<5 ms
5	LLM response	GPT-4o-mini (OpenAI)	300–600 ms
6	Text-to-Speech	ElevenLabs WebSocket streaming	110–155 ms TTFB
7	Audio playback	μ-law → EnableX WebSocket	real-time
8	Session state	Redis (Railway)	<5 ms

Table 2 — End-to-end call flow with observed latencies

2.2 Key Technical Parameters

- Audio codec: G.711 μ -law, 8 kHz, 8-bit (160 bytes = 20 ms per frame)
- TTS output: ulaw_8000 streamed over ElevenLabs WebSocket (stream-input endpoint)
- STT input: resampled to 16 kHz PCM WAV, sent as batch REST to Sarvam
- Barge-in: speculative STT fires at 8 frames before VAD silence confirmation
- LLM context: last 12 turns, KB injected as system prompt dynamic variable
- Language switching: automatic per-turn (Hindi $\leftrightarrow$ English $\leftrightarrow$ Marathi etc.)

2.3 Infrastructure

- Orchestrator: Railway (Southeast Asia), Node 20 Alpine, pm2-runtime
- Dashboard: Vercel (static), proxies /call/* $\rightarrow$ Railway orchestrator
- Database: Railway Redis (session state, recording cache)
- Telephony: EnableX (DID +91 number, outbound)
- CI/CD: GitHub $\rightarrow$ Railway auto-deploy on master push

3. Cost Analysis

3.1 Actual Usage (May 14 – June 5, 2026)

ElevenLabs API usage CSV was reviewed. The system went live on May 14, 2026. Total verified volume is small — pilot phase only.

Service	Volume	Period	Est. Cost
ElevenLabs TTS (ulaw_8000)	~1,170 calls	May 14 – June 5	~\$15–25
ElevenLabs STT (Scribe)	~667 calls	May 14 – June 5	~\$5–10
Sarvam Saarika v2.5 STT	production primary	ongoing	~■0.005/req
OpenAI GPT-4o-mini	~1 call / turn	ongoing	~\$0.0004/turn
EnableX telephony	per-minute billing	ongoing	~■1.5–2/min
Railway hosting	flat subscription	ongoing	~\$20/mo

Table 3 — Actual API usage since go-live

3.2 Per-Minute Cost Breakdown (Production Scale)

At 1,000 minutes/month (realistic production volume), the blended per-minute cost is estimated below. ElevenLabs dominates at scale despite good unit economics at pilot volume.

Component	Provider	Unit Cost	Per Min (1k min/mo)	% of Total
TTS (streaming)	ElevenLabs	\$0.30/1k chars	■1.80–2.40	~50%
STT (batch REST)	Sarvam Saarika	■0.005/req	■0.30–0.50	~10%
LLM	GPT-4o-mini	\$0.40/1M tokens	■0.20–0.40	~8%
Telephony	EnableX	■1.5–2/min DID	■1.50–2.00	~45%
Infrastructure	Railway	~\$20/mo flat	■0.03	<1%
TOTAL			■3.80–5.30/min	100%

Table 4 — Blended per-minute cost at production scale (1k min/mo)

3.3 Competitor Comparison

Provider	Model	Cost (API)	Cost (White-label)	Latency Claim
Ravan.ai (Agni)	Proprietary + Sarvam?	■4/min	■8/min	<300 ms
Prophunt (current)	EL + Sarvam + GPT-4o-mini	■3.8–5.3/min	—	~1.8s P50
Prophunt (Phase 1)	Sarvam Bulbul V3 + Sarvam STT	~■2.8–3.5/min	■5–6/min target	~1.5s P50 est.
Prophunt (Phase 3)	+ Gemini 2.5 Flash LLM	~■2.2–2.8/min	■5/min target	<1.2s P50 est.

Table 5 — Competitor cost comparison (Ravan.ai reference pricing as of June 2026)

4. Latency Analysis

Total turn latency (caller finishes speaking → agent audio begins) is dominated by Sarvam batch STT. ElevenLabs TTS TTFB is excellent and is **not** the bottleneck.

4.1 Latency Waterfall (per turn)

Stage	P50	P90	Bottleneck?	Fix Available
VAD silence detection	~400 ms	~600 ms	Minor	Speculative STT (done)
Sarvam STT (batch REST)	900–1 200 ms	1 600–2 025 ms	MAJOR ✗	WebSocket streaming STT
GPT-4o-mini LLM	300–500 ms	600–900 ms	Moderate	Gemini Flash (Phase 3)
ElevenLabs TTS TTFB	110–155 ms	200–280 ms	None ✓	N/A (already fast)
Audio playout start	~20 ms	~20 ms	None ✓	N/A
TOTAL turn latency	~1.8 s	~2.5–3.5 s		

Table 6 — Per-turn latency breakdown with observed P50/P90 values

4.2 ElevenLabs TTS — Detailed Metrics (from API usage CSV)

ElevenLabs TTS performance is excellent. TTFB of 110–155 ms median is well within acceptable bounds for telephony. ElevenLabs is NOT the latency problem.

Metric	Observed Value	Industry Target	Assessment
TTFB P50	110–155 ms	<300 ms	✓ Excellent
TTFB P90	200–280 ms	<500 ms	✓ Excellent
Total TTS calls (lifetime)	~1,170	—	Pilot scale
STT calls via ElevenLabs Scribe	~667	—	Secondary STT
First call date	May 14, 2026	—	~3 weeks live
Total ElevenLabs spend	~\$15–35	—	Negligible

Table 7 — ElevenLabs usage metrics from API CSV (May 14 – June 5, 2026)

5. Bugs Found and Fixed (June 4–5, 2026)

Two critical bugs were identified and fixed during this audit session. Both are now live on GitHub master and will be deployed by Railway automatically.

5.1 Bug: Audio Crackling on Every Utterance

Severity: High — affects every single agent utterance

Every agent utterance ended with a loud click or pop, making calls sound unprofessional.

Root Cause

G.711 μ -law audio silence is encoded as 0xFF (decodes to 0). The code was padding leftover audio frames with `Buffer.alloc(N)`, which fills with 0x00. In μ -law encoding, 0x00 is the *maximum negative amplitude* (-32,124), not silence. This fired a sharp negative spike at the end of every utterance.

Fix Applied

```
// Before (BUG): 0x00 = max amplitude in  $\mu$ -law = LOUD CLICK
Buffer.alloc(160 - (leftover.length % 160))

// After (FIX): 0xFF = silence in  $\mu$ -law (decodes to 0)
Buffer.alloc(160 - (leftover.length % 160), 0xff)
```

Additionally, `optimize_streaming_latency` was lowered from 3 $\rightarrow$ 2 (level 3 causes additional audio artifacts on ulaw_8000 telephony).

5.2 Bug: Orchestrator Crash $\rightarrow$ HTTP 502 on Every Call Attempt

Severity: Critical — blocks all outbound calls

All outbound call attempts were failing instantly with Railway's error: `{"status":"error","code":502,"message":"Application failed to respond"}`

Root Cause (Primary)

The crackling-fix commits from the previous session used the GitHub MCP `create_or_update_file` tool, which silently truncated the 208 KB file to 14 lines (only require statements). The orchestrator container was executing a file with no server, no routes, and no `server.listen()` call — Node.js exited immediately after parsing, causing an instant 502 on every request.

Root Cause (Secondary)

No Redis error handler was registered. With `ioredis lazyConnect: false`, connection failures emit 'error' events on the client object. Without a listener, Node.js EventEmitter throws the error as an `uncaughtException`, crashing the process.

Fix Applied

- Restored the full 4,280-line orchestrator file via PR merge (bypassed the truncating MCP tool)
- Added `redis.on('error', ...)` handler to prevent ioredis connection errors from crashing Node
- Added `process.on('uncaughtException', ...)` and `process.on('unhandledRejection', ...)` handlers
- Wrapped entire `/call/dial` handler in a single try/catch covering all pre-dial steps
- Added diagnostic log before `EnableX` call for easier Railway log debugging

File	Change	Commit
backend/orchestrator/main.js	Redis error handler	800914a
backend/orchestrator/main.js	Process-level crash guards	800914a

File	Change	Commit
backend/orchestrator/main.js	/call/dial try/catch	800914a
backend/orchestrator/main.js	μ-law 0xFF silence padding	800914a
backend/orchestrator/main.js	ElevenLabs latency 3→2	800914a
backend/orchestrator/main.js	Restore full file (merge)	d9f0bcb

Table 8 — Fixes applied to production codebase, June 5, 2026

6. Optimisation Roadmap

Three phases are recommended in priority order. Phase 1 delivers the largest cost reduction. Phase 2 is the biggest latency improvement. Phase 3 completes the LLM cost optimisation.

Phase	Change	Cost Impact	Latency Impact	Effort	Priority
1	ElevenLabs TTS → Sarvam Bulbul V3	■1.5–2/min savings (~45% TTS cost cut)	Neutral (same TTFB)	Medium 2–3 days	HIGH
2	Sarvam batch STT → WebSocket streaming STT	Minimal cost change	–700–1 200 ms P50 (biggest win)	Medium 3–5 days	HIGH
3	GPT-4o-mini → Gemini 2.5 Flash	~40% LLM cost cut better Hindi-English mix	–100–200 ms P50	Low 1–2 days	MEDIUM

Table 9 — Three-phase optimisation roadmap

6.1 Phase 1 — Sarvam Bulbul V3 TTS (Recommended Next Step)

Sarvam Bulbul V3 is an Indian-language TTS model with multi-language support including Hindi, Marathi, Tamil, Telugu, Punjabi, Bengali and English. At production scale it costs roughly 60–70% less than ElevenLabs for the same audio.

- API: REST POST with text + language + speaker parameters
- Output: WAV/PCM — needs μ -law transcoding before EnableX (one ffmpeg call)
- Voices: 10+ Indian-language speakers, comparable expressiveness to ElevenLabs for Hindi
- ElevenLabs stays as fallback for English calls where accent quality matters
- Estimated savings at 1k min/mo: ■1,500–2,000/mo

6.2 Phase 2 — Sarvam WebSocket Streaming STT

The current Sarvam Saarika v2.5 batch REST STT adds 900–2,025 ms per turn. Sarvam's WebSocket streaming STT delivers partial transcripts as the caller speaks, cutting effective latency to ~200–400 ms (barge-in on partial).

- Protocol: WebSocket, 8 kHz PCM 16-bit (same format as EnableX audio — no resampling needed)
- Output: Partial transcripts → final transcript on silence detection
- Integration: replaces `runSarvamSTT()` REST call; VAD triggers send-end message
- LLM can start processing on partial transcript (speculative processing)
- This is the single highest-latency-impact change available

6.3 Phase 3 — Gemini 2.5 Flash LLM

- Cost: ~40% cheaper than GPT-4o-mini for same token volume
- Quality: Better at Hindi-English code-mixing (Hinglish), stronger at real estate vocabulary
- Latency: 100–200 ms faster TTFT on average
- Risk: low — swap the API call in `callLLM()`, keep the same prompt structure

7. Open Risks & Recommendations

Risk	Severity	Recommendation	Status
Railway SE Asia storage incident (June 4–5)	High	Monitor — resolved per Railway status page	Resolved
main.js file truncation via MCP tool	Critical	Always push large files via git push, never create <code>FixedUpdate</code> file for >10KB files	Fixed
No Redis error handler	High	Added <code>redis.on('error')</code> — prevents crash on blip	Fixed June 5
No process-level crash handlers	Medium	Added <code>uncaughtException</code> + <code>unhandledRejection</code>	Fixed June 5
ElevenLabs single point of failure	Medium	Add Sarvam Bulbul V3 as fallback TTS	Phase 1
Sarvam batch STT latency (900–2025ms)	High	Migrate to WebSocket streaming STT	Phase 2
No call recording on Railway volume reset	Medium	Redis-based recording cache already in place	Monitored
EnableX DID number validity	Medium	Verify <code>ENABLEX_FROM_NUMBER</code> env var after <code>Manyim</code> incident	Monitor

Table 10 — Risk register with current status

Appendix A — Environment Variables Checklist

These environment variables must be correctly set in Railway for the orchestrator to function. Verify after any infrastructure incident.

Variable	Purpose	Required?	Notes
ENABLEX_APP_ID	EnableX API authentication	Critical	Verify after incident
ENABLEX_APP_KEY	EnableX API authentication	Critical	Verify after incident
ENABLEX_FROM_NUMBER	Outbound caller DID	Critical	Must be active number
REDIS_URL	Session state store	Critical	Set by Railway automatically
OPENAI_API_KEY	GPT-4o-mini LLM	Critical	Check quota/billing
ELEVENLABS_API_KEY	TTS + STT (Scribe)	Critical	Check credits
ELEVENLABS_VOICE_FEMALE	Female voice ID	Required	Defaults to Priya
ELEVENLABS_VOICE_MALE	Male voice ID	Optional	Defaults to Adam
SARVAM_API_KEY	Saarika v2.5 STT	Critical	Primary STT provider
PUBLIC_HOST	Self-URL for EnableX callbacks	Critical	Railway public URL
ORCHESTRATOR_INTERNAL_TOKEN	Inter-service auth	Required	
RECORDINGS_DIR	WAV file storage path	Optional	Defaults to /data/recordings

Table 11 — Railway environment variables checklist

Appendix B — Ravan.ai Agni — Known Facts vs. Inferences

Ravan.ai's internal architecture is not publicly documented. The following distinguishes confirmed facts from educated inferences.

Claim	Source	Confidence
Ravan.ai charges █4/min API	Ravan.ai public pricing page	Confirmed
Ravan.ai charges █8/min white-label	Ravan.ai public pricing page	Confirmed
Ravan.ai Agni claims <300 ms latency	Ravan.ai marketing material	Unverified
Ravan.ai uses Sarvam for STT/TTS	Inference (Indian language specialisation)	Unconfirmed
Ravan.ai uses Bulbul V2/V3 TTS	Inference based on language coverage	Unconfirmed
Ravan.ai uses EnableX for telephony	Unknown — could be Twilio/Plivo/Exotel	Unknown
Ravan.ai has 50+ language support	Ravan.ai product page	Confirmed

Table 12 — Ravan.ai intelligence assessment